NN
ACADEMY

All of IGCSE Maths in 10 Minutes

Whole-syllabus question deck — Cambridge IGCSE Mathematics 0580, Extended. The 10-minute cut: every topic that actually costs marks, one question per page, worked live on screen. Blank workspace sized for annotation in Xournal++.

39	53 / 72	9:59	16 : 9
questions	Extended subtopics	target runtime	254 × 143 mm

PREPARED FOR	DATE	SYLLABUS	EXAM SERIES
Nur Nabi	**21 September 2026**	**0580 (2025–2027) · Extended**	**2026 & 2027**

READ THIS BEFORE YOU RECORD

How to use this deck

In Xournal++

- Open this PDF as the **background** (Annotate PDF), not as an image import — the text stays crisp at any zoom.
- Page size is **254 × 143 mm**, exactly 16:9, so a full-page view fills a 1920 × 1080 recording frame with no letterboxing.
- The faint dot grid is 5 mm and is there to keep your handwriting level. It is light enough to disappear under video compression.
- Set the pen to roughly **1.5 mm**, dark blue or black. Use red only for the final answer so the freeze-frame reads at a glance.
- Export the annotated file afterwards as the downloadable "worked solutions" PDF — that is your description link and your email-list magnet.

Pacing

- Each page carries its own time budget in the top-right. It is a ceiling, not a target.
- If a page overruns, cut the explanation, never the worked answer.
- Hard cut between pages. No transitions and no "now let's look at" — there is no runtime for connective tissue.

Everything here is original. The questions were written for this deck against the published syllabus. Cambridge past-paper questions are copyright and must not be screen-recorded or retyped — these are fresh questions in the same style, which is why you can publish and monetise the video safely. The whole-syllabus sweep format is a genre, not anyone's property; the wording, the numbers and the working are yours.

Say this at the end, and put it in the description. A 10-minute sweep is a revision aid and a gap-finder, not a course. Students who rely on it alone will be short of practice. Four seconds of honesty buys you the comment section.

SEGMENT MAP

The 10-minute running order

> Paste the **Start** and **Segment** columns into the YouTube description as chapter markers. The **Budget** column is your edit discipline, and at this length there is no slack in it at all.

START	SEGMENT	CODES	BUDGET
0:00	**Intro — the promise**	–	0:30
0:30	**Number** 10 questions	E1.1 - E1.18	2:04
2:34	**Algebra and graphs** 11 questions	E2.1 - E2.13	2:16
4:50	**Coordinate geometry** 1 questions	E3.1 - E3.7	0:24
5:14	**Geometry** 3 questions	E4.1 - E4.8	0:46
6:00	**Mensuration** 3 questions	E5.1 - E5.5	0:42
6:42	**Trigonometry** 4 questions	E6.1 - E6.6	1:02
7:44	**Transformations and vectors** 2 questions	E7.1 - E7.4	0:44
8:28	**Probability** 2 questions	E8.1 - E8.4	0:24
8:52	**Statistics** 3 questions	E9.1 - E9.7	0:42
9:34	**Outro — how to use this**	–	0:25
9:59	**Total runtime**		

1

Number

E1.1 - E1.18 · 10 questions · 2:04 on the clock

Question 01

Types of number, HCF and LCM

Also know: natural numbers · integers · primes · squares · cubes · rational and irrational · reciprocals

(a) Write 168 and 180 each as a product of its prime factors.

$168 = 8 \times 21 = 2^3 \times 3 \times 7$

$180 = 4 \times 45 = 2^2 \times 3^2 \times 5$

$2^3 \times 3 \times 7$, $2^2 \times 3^2 \times 5$ [2]

(b) Hence find the highest common factor and the lowest common multiple of 168 and 180.

HCF: common primes, lowest powers → $2^2 \times 3$

LCM: every prime, highest powers → $2^3 \times 3^2 \times 5 \times 7$

HCF = 12, LCM = 2520 [2]

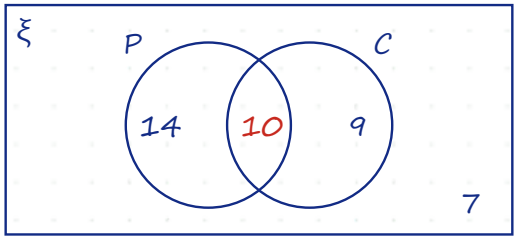
Question 02

Sets and Venn diagrams

In a group of 40 students, 24 study Physics, 19 study Chemistry and 7 study neither.

Notation: ξ $\cup$ $\cap$ A' $\in$ $\notin$ $\subseteq$ $\varnothing$ $n(A)$

(a) Complete the Venn diagram.



$P \cup C = 40 - 7 = 33$
$24 + 19 - x = 33$
$43 - x = 33 \rightarrow x = 10$
P only $= 14$, C only $= 9$

[2]

(b) Find n($P \cap C$).

$P \cap C$ = in both

10

[1]

Question 02

Sets and Venn diagrams

In a group of 40 students, 24 study Physics, 19 study Chemistry and 7 study neither.

(c) Find n($P \cap C'$).

P but not C = 24 - 10

14 ..... [1]

Question 03

Powers, roots and the index laws

Recall squares to 15^2 and cubes of 1, 2, 3, 4, 5 and 10

(a) Work out $27^{-2/3}$, giving your answer as a fraction.

$27^{-2/3} = \frac{1}{27^{2/3}}$

$= \frac{1}{(\sqrt[3]{27})^2} = \frac{1}{3^2}$

Index laws

$x^0 = 1$ $\quad x^{-n} = \frac{1}{x^n}$

$x^m \times x^n = x^{m+n}$ $\quad x^m \div x^n = x^{m-n}$

$(x^m)^n = x^{mn}$ $\quad x^{m/n} = (\sqrt[n]{x})^m$

$\frac{1}{9}$ [2]

(b) Simplify $(2a^3b)^4 \div 4a^5b^2$.

$(2a^3b)^4 = 16a^{12}b^4$

$16a^{12}b^4 \div 4a^5b^2 = 4a^{12-5}b^{4-2}$

$4a^7b^2$ [2]

Question 04

Fractions, decimals and recurring decimals

(a) Write $0.4\dot{2}$ as a fraction in its simplest form.

$x = 0.4242\ldots \quad 100x = 42.4242\ldots$

$99x = 42 \rightarrow x = \frac{42}{99}$

$\frac{14}{33}$ [2]

(b) Work out $2\frac{1}{3} - 1\frac{3}{4}$, giving your answer as a fraction in its simplest form.

$\frac{7}{3} - \frac{7}{4} = \frac{28}{12} - \frac{21}{12}$

$\frac{7}{12}$ [2]

Question 05

Standard form

Answers must be in the form $A \times 10^n$ with $1 \leq A < 10$

(a) Work out $(3.2 \times 10^5) \div (8 \times 10^{-2})$.

$(3.2 \div 8) \times 10^{5-(-2)} = 0.4 \times 10^7$

4×10^6 [2]

(b) Work out $(6 \times 10^4) + (4.5 \times 10^5)$.

$60\,000 + 450\,000 = 510\,000$

5.1×10^5 [2]

Question 06

Limits of accuracy (bounds)

A rectangle measures 12.4 cm by 8.6 cm, each correct to 1 decimal place.

(a) Write down the lower and upper bound of each side.

1 d.p. → ± 0.05

$12.35 \le L < 12.45, \quad 8.55 \le W < 8.65$

12.35, 12.45 ; 8.55, 8.65 [2]

(b) Calculate the upper bound of the area.

UB area = UB × UB = 12.45 × 8.65

107.6925 cm^2 [2]

(c) Calculate the lower bound of the perimeter.

LB perimeter = 2(12.35 + 8.55)

41.8 cm [2]

Question 07

Ratio and proportion

An amount of money is shared between Amina, Bilal and Chowdhury in the ratio 3 : 5 : 7.

Chowdhury receives $84 more than Amina. Find the total amount of money shared.

WORKING

C − A = 7 − 3 = 4 parts
4 parts = $84 → 1 part = $21
total = 3 + 5 + 7 = 15 parts
15 × 21

$315 [3]

Question 08

Percentages and reverse percentages

(a) After a 15% increase, the price of a phone is $161. Calculate the price before the increase.

100% + 15% = 115% = $161
161 ÷ 1.15 (not 161 × 0.85)

$140 ..... [2]

(b) $2000 is invested at 3.5% per year compound interest. Find the value after 5 years.

multiplier 1.035, 5 years
$2000 \times 1.035^5 = 2375.372...$

$2375.37 ..... [3]

Question 09

Exponential growth and decay

A car is bought for $18 000. Its value depreciates by 12% each year.

Calculate its value after 4 years, correct to the nearest dollar.

Growth: $P(1 + r)^n$ · Decay: $P(1 - r)^n$ · knowledge of e is not required

WORKING

decay → multiplier $1 - 0.12 = 0.88$

$18\,000 \times 0.88^4$

$= 10\,794.52\ldots$

$10 795 [3]

Question 10

Surds

(a) Simplify $\sqrt{50} + \sqrt{18}$.

$\sqrt{50} = \sqrt{25} \times \sqrt{2} = 5\sqrt{2}, \quad \sqrt{18} = \sqrt{9} \times \sqrt{2} = 3\sqrt{2}$

$8\sqrt{2}$ [2]

(b) Simplify $6 \div \sqrt{3}$.

$\frac{6}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{6\sqrt{3}}{3}$

$2\sqrt{3}$ [2]

(c) Rationalise the denominator of $1 \div (2 + \sqrt{3})$.

$\frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{4-3}$

$2 - \sqrt{3}$ [2]

2

Algebra and graphs

E2.1 - E2.13 · 11 questions · 2:16 on the clock

Question 11

Expanding and factorising

(a) Expand and simplify $(2x - 3)(x + 5)$.

$2x^2 + 10x - 3x - 15$

$2x^2 + 7x - 15$ [2]

(b) Factorise $6x^2 - x - 12$.

$6 \times (-12) = -72 \rightarrow -9$ and $+8$

$6x^2 - 9x + 8x - 12 = 3x(2x - 3) + 4(2x - 3)$

$(2x - 3)(3x + 4)$ [2]

Question 11

Expanding and factorising

(c) Factorise $9x^2 - 49y^2$.

difference of two squares:
$(3x)^2 - (7y)^2$

$(3x - 7y)(3x + 7y)$ [2]

(d) Factorise by grouping: $ax + ay - 2bx - 2by$.

$a(x + y) - 2b(x + y)$

$(a - 2b)(x + y)$ [2]

Question 12

Algebraic fractions

(a) Simplify $(x^2 - 9) \div (x^2 + 7x + 12)$.

$$\frac{(x - 3)(x + 3)}{(x + 3)(x + 4)}$$

$$\frac{x - 3}{x + 4}$$ [3]

(b) Write as a single fraction $3/(x + 1) + 2/(x - 2)$.

$$\frac{3(x - 2) + 2(x + 1)}{(x + 1)(x - 2)} = \frac{3x - 6 + 2x + 2}{(x + 1)(x - 2)}$$

$$\frac{5x - 4}{(x + 1)(x - 2)}$$ [3]

Question 13

Equations and rearranging

(a) Solve simultaneously $3x + 2y = 16$ and $x - y = 2$.

$\times 2$: $2x - 2y = 4$
add: $5x = 20 \rightarrow x = 4$
$4 - y = 2 \rightarrow y = 2$

$x = 4,\ y = 2$ [3]

(b) Solve $2x^2 - 7x - 3 = 0$, giving answers correct to 2 decimal places.

$$x = \frac{7 \pm \sqrt{(49 + 24)}}{4} = \frac{7 \pm \sqrt{73}}{4}$$

$x = 3.89$ or -0.39 [4]

Question 13

Equations and rearranging

(c) Make r the subject of $V = \frac{2}{3}\pi r^3$.

$3V = 2\pi r^3$

$r^3 = \frac{3V}{2\pi}$

$r = \sqrt[3]{\frac{3V}{2\pi}}$ [3]

(d) Write $x^2 - 6x + 2$ in the form $(x - a)^2 - b$.

half of -6 is -3

$(x - 3)^2 - 9 + 2$

$(x - 3)^2 - 7$ [2]

Question 14

Non-linear simultaneous equations

Solve simultaneously

$$y = x^2 - 4 \quad \text{and} \quad y = 3x$$

WORKING

$x^2 - 4 = 3x$

$x^2 - 3x - 4 = 0$

$(x - 4)(x + 1) = 0$

$x = 4 \rightarrow y = 12$

$x = -1 \rightarrow y = -3$

$x = 4, y = 12$ or $x = -1, y = -3$

Question 15

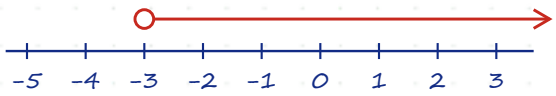
Inequalities

Dividing by a negative reverses the inequality sign

(a) Solve $5 - 2x < 11$ and show the solution on a number line.

$-2x < 6$

$\div (-2) \rightarrow$ flip the sign



open circle: -3 not included

$x > -3$ [3]

(b) Solve $-3 \leq 2x + 1 < 9$.

$-4 \leq 2x < 8$

$\div 2$ all three parts

$-2 \leq x < 4$ [2]

Question 16

Sequences

Find the *n*th term of each sequence.

Linear · quadratic · cubic · exponential · and simple combinations

(a) 7, 11, 15, 19, ...

+4 each time → 4n
4(1) = 4 → 7 − 4 = 3

4n + 3 [2]

(b) 3, 8, 15, 24, 35, ...

diffs 5, 7, 9, 11 → 2nd diff 2 → n^2
3, 8, 15, 24 − (1, 4, 9, 16) = 2, 4, 6, 8 → 2n

$n^2 + 2n$ [2]

Question 16

Sequences

Find the *n*th term of each sequence.

(c) 5, 10, 20, 40, ...

× 2 each time → geometric
first term 5

$5 \times 2^{n-1}$ [2]

Question 17

Direct and inverse proportion

y is inversely proportional to x^2. When $x = 2$, $y = 12$.

(a) Find y in terms of x.

$y = \frac{k}{x^2}$ $\quad 12 = \frac{k}{4} \rightarrow k = 48$

$y = \frac{48}{x^2}$ [2]

(b) Find y when $x = 4$.

$y = \frac{48}{4^2} = \frac{48}{16}$

$y = 3$ [1]

(c) Find x when $y = 0.75$.

$0.75 = \frac{48}{x^2} \rightarrow x^2 = 64$

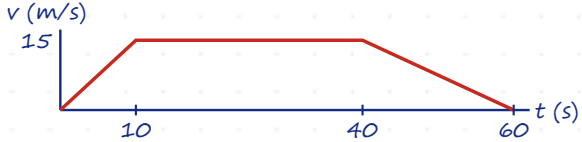
$x = \pm 8$ [2]

Question 18

Graphs in practical situations

A car accelerates uniformly from rest to 15 m/s in 10 s, travels at 15 m/s for 30 s, then decelerates uniformly to rest in 20 s.

(a) Sketch the speed–time graph.



up for 10 s, flat for 30 s,
down for 20 s

[2]

(b) Find the acceleration in the first 10 s.

$$\text{acceleration} = \text{gradient} = \frac{15}{10}$$

$1.5\ \text{m/s}^2$ [1]

(c) Find the total distance travelled.

$$\text{distance} = \text{area under graph}$$

$$\frac{1}{2}(10)(15) + 30(15) + \frac{1}{2}(20)(15) = 75 + 450 + 150$$

675 m [3]

Question 19

Sketching curves

Know the shapes: linear · quadratic · cubic · reciprocal · exponential

(a) Write $y = x^2 - 6x + 5$ in completed-square form and state the turning point and the roots.

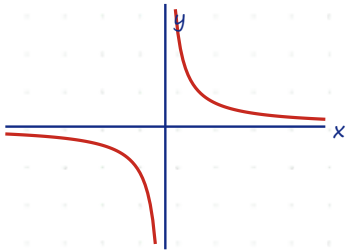
$(x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$

turning point $(3, -4)$

$(x - 3)^2 = 4 \rightarrow x - 3 = \pm 2$

$(3, -4)$; $x = 1, x = 5$ [4]

(b) Sketch $y = 4/x$ and state the equations of its asymptotes.



never touches the axes

$x = 0 \rightarrow$ undefined

$x = 0$ and $y = 0$ [3]

Question 20

Differentiation

$y = x^3 - 6x^2 + 5$

(a) Find dy/dx.

bring the power down, reduce by 1

Differentiation

$$y = ax^n \rightarrow \frac{dy}{dx} = anx^{n-1}$$

$3x^2 - 12x$ [2]

(b) Find the coordinates of the two stationary points.

$3x^2 - 12x = 0 \rightarrow 3x(x - 4) = 0$

$x = 0$: $y = 5$; $x = 4$: $y = 64 - 96 + 5 = -27$

(0, 5) and (4, -27) [4]

(c) Determine which is a maximum and which is a minimum.

$\frac{d^2y}{dx^2} = 6x - 12$

$x = 0 \rightarrow -12 < 0$ max; $x = 4 \rightarrow 12 > 0$ min

(0, 5) max, (4, -27) min [3]

Question 21

Functions

$f(x) = 3x - 4$ $g(x) = x^2$

(a) Find fg(2).

$g(2) = 4 \rightarrow f(4) = 12 - 4$

8 [1]

(b) Find gf(2).

$f(2) = 2 \rightarrow g(2) = 2^2$

4 [1]

Question 21

Functions

$f(x) = 3x - 4 \quad g(x) = x^2$

(c) Find $f^{-1}(x)$.

$y = 3x - 4 \rightarrow x = \frac{y + 4}{3}$

swap x and y

$f^{-1}(x) = \frac{x + 4}{3}$ [2]

(d) Find $ff(x)$ in its simplest form.

$f(3x - 4) = 3(3x - 4) - 4$

$9x - 16$ [2]

3

Coordinate geometry

E3.1 - E3.7 · 1 questions · 0:24 on the clock

Question 22

All of coordinate geometry in one pair of points

A is the point $(-2, 1)$ and B is the point $(6, 7)$.

One pair of points answers all seven subtopics — that is the trick that makes this segment fit

(a) Midpoint of AB.

$$\left(\frac{-2+6}{2}, \frac{1+7}{2}\right)$$

(2, 4) [1]

(b) Length of AB.

$$\sqrt{(8^2 + 6^2)} = \sqrt{100}$$

10 [2]

Question 22

All of coordinate geometry in one pair of points

A is the point $(-2, 1)$ and B is the point $(6, 7)$.

(c) Gradient of AB.

$$\frac{7-1}{6-(-2)} = \frac{6}{8}$$

$\frac{3}{4}$ [1]

(d) Equation of AB.

$$y = \frac{3}{4}x + c, \quad 7 = \frac{3}{4}(6) + c \rightarrow c = 2.5$$

$y = \frac{3}{4}x + 2.5$ [2]

Question 22

All of coordinate geometry in one pair of points

A is the point $(-2, 1)$ and B is the point $(6, 7)$.

(e) Equation of the line through the origin parallel to AB.

parallel → same gradient; origin → $c = 0$

$y = \frac{3}{4}x$

[1]

(f) Equation of the line through B perpendicular to AB.

perpendicular gradient $= -\frac{4}{3}$

$7 = -\frac{4}{3}(6) + c \rightarrow c = 15$

$y = -\frac{4}{3}x + 15$

[3]

4

Geometry

E4.1 - E4.8 · 3 questions · 0:46 on the clock

Question 23

Geometrical terms and angles

(a) For a regular 12-sided polygon, find the size of each exterior angle, each interior angle, and the sum of the interior angles.

exterior = 360 ÷ 12 = 30°
interior = 180 − 30 = 150°
sum = (12 − 2) × 180

30°, 150°, 1800° [3]

(b) Two parallel lines are cut by a transversal. Name the angle relationships you would use and give a reason for each.

corresponding angles are equal
alternate angles are equal
co-interior angles add up to 180°

[3]

Question 24

Similarity and symmetry

Two similar cones have heights 4 cm and 6 cm. The smaller cone has volume 32 cm^3 and surface area 20 cm^2.

Length factor k → area factor k^2 → volume factor k^3

(a) Find the volume of the larger cone.

length factor $k = \frac{6}{4} = 1.5$

volume × k^3: $32 \times 1.5^3 = 32 \times 3.375$

108 cm^3 [2]

(b) Find its surface area.

area × k^2: $20 \times 1.5^2 = 20 \times 2.25$

45 cm^2 [2]

Question 25

Circle theorems

Find the lettered angles, giving a geometrical reason for each.

Semicircle · tangent ⊥ radius · centre = 2 × circumference · same segment · cyclic quadrilateral · alternate segment · equal chords · tangents from a point are equal

(a) Angle at the centre is 130° — find the angle at the circumference.

130 ÷ 2
angle at centre = 2 × angle at circumference

65° ..... [2]

(b) Hence the opposite angle of the cyclic quadrilateral.

180 − 65
opposite angles of a cyclic quadrilateral add to 180°

115° ..... [2]

Question 25

Circle theorems

Find the lettered angles, giving a geometrical reason for each.

(c) A tangent meets a chord at 58° — find the angle in the alternate segment.

alternate segment theorem

58° ..... [1]

(d) *OP* = 17 cm, radius 8 cm — find the tangent length *PT*.

tangent ⊥ radius → right angle at T

PT = √(17² − 8²) = √225

15 cm ..... [3]

5

Mensuration

E5.1 - E5.5 · 3 questions · 0:42 on the clock

Question 26

Area, perimeter and compound shapes

(a) A trapezium has parallel sides 7 cm and 11 cm and perpendicular height 6 cm. Find its area.

$\frac{1}{2}(a + b)h = \frac{1}{2}(7 + 11) \times 6$

54 cm² ..... [2]

(b) A rectangle 10 cm by 6 cm has a semicircle of diameter 6 cm removed from one end. Find the area remaining.

$10 \times 6 - \frac{1}{2}\pi(3)^2 = 60 - 14.14$

45.9 cm² ..... [3]

Question 27

Circles, arcs and sectors

A sector of a circle has radius 9 cm and angle 140°.

(a) Find the arc length.

$\frac{140}{360} \times 2\pi \times 9 = 21.99$

Sectors

$\text{arc} = \frac{\theta}{360} \times 2\pi r \quad \text{area} = \frac{\theta}{360} \times \pi r^2$

22.0 cm [2]

(b) Find the area of the sector.

$\frac{140}{360} \times \pi \times 9^2 = 98.96$

99.0 cm^2 [2]

(c) Find the perimeter of the sector.

arc + 2 radii = 21.99 + 9 + 9

40.0 cm [1]

Question 28

Surface area and volume

Cone: $\frac{1}{3}\pi r^2 h$ and curved area $\pi r l$ · Sphere: $\frac{4}{3}\pi r^3$ and $4\pi r^2$ — both given in the formula list

(a) A cone has radius 5 cm and vertical height 12 cm. Find its slant height, its total surface area and its volume.

$l = \sqrt{(5^2 + 12^2)} = 13$

$SA = \pi r l + \pi r^2 = 65\pi + 25\pi = 90\pi$

$V = \frac{1}{3}\pi(5^2)(12) = 100\pi$

13 cm; 283 cm^2; 314 cm^3 [5]

(b) A sphere has radius 6 cm. Find its volume and its surface area.

$V = \frac{4}{3}\pi(6^3) = 288\pi$

$SA = 4\pi(6^2) = 144\pi$

905 cm^3; 452 cm^2 [3]

6

Trigonometry

E6.1 - E6.6 · 4 questions · 1:02 on the clock

Question 29

Pythagoras and right-angled trigonometry

In a right-angled triangle the side opposite angle θ is 7 cm and the hypotenuse is 12 cm.

SOH · CAH · TOA

(a) Find θ.

O and H → SOH: $\sin\theta = \frac{7}{12}$

$\theta = \sin^{-1}\left(\frac{7}{12}\right)$

$\theta = 35.7°$ [2]

(b) Find the third side.

$\sqrt{(12^2 - 7^2)} = \sqrt{95}$

9.75 cm [2]

Question 30

Trigonometric graphs and equations

Solve for $0° \leq x \leq 360°$.

Sketch the curve first — it tells you how many solutions there are

(a) $\sin x = 0.5$

$\sin^{-1}(0.5) = 30°$
sine symmetry: $180 - 30 = 150°$

$x = 30°, 150°$ [2]

(b) $2\cos x + 1 = 0$

$\cos x = -0.5$
$\cos^{-1}(-0.5) = 120°, \quad 360 - 120 = 240°$

$x = 120°, 240°$ [3]

Question 31

Sine rule, cosine rule and area

Two sides and the angle between → cosine rule. Anything else → sine rule. Both are in the formula list.

(a) In triangle ABC, $A = 47°$, $B = 63°$, $a = 8.4$ cm. Find b.

$$\frac{b}{\sin 63} = \frac{8.4}{\sin 47} \rightarrow b = \frac{8.4 \sin 63}{\sin 47} = 10.23$$

10.2 cm ..... [3]

(b) In a different triangle, $b = 8$ cm, $c = 11$ cm, $A = 47°$. Find a.

$$a^2 = 8^2 + 11^2 - 2(8)(11)\cos 47 = 64.97$$

8.06 cm ..... [3]

(c) Find its area.

$$\frac{1}{2}bc \sin A = \frac{1}{2}(8)(11) \sin 47$$

32.2 cm^2 ..... [2]

Question 32

Pythagoras and trigonometry in 3D

A cuboid has a rectangular base 6 cm by 8 cm and height 10 cm.

(a) Find the length of the space diagonal.

$\sqrt{(6^2 + 8^2 + 10^2)} = \sqrt{200}$

14.1 cm ..... [2]

(b) Find the angle between the space diagonal and the base.

base diagonal $= \sqrt{(6^2 + 8^2)} = 10$

$\tan\theta = \frac{10}{10} = 1$

45° ..... [3]

7

Transformations and vectors

E7.1 - E7.4 · 2 questions · 0:44 on the clock

Question 33

Transformations

Describe each transformation fully.

Reflection: the mirror line · Rotation: angle, direction, centre · Enlargement: scale factor, centre · Translation: the column vector

(a) A reflection.

Reflection + the mirror line as an equation
e.g. Reflection in the line $y = x$

..... [2]

(b) A rotation.

Rotation + angle + direction + centre
e.g. Rotation 90° clockwise about (0, 0)

..... [3]

Question 33

Transformations

Describe each transformation fully.

(c) An enlargement with scale factor −2, centre (1, 1).

Enlargement, scale factor −2, centre (1, 1)
negative → image on the other side of the centre, upside down
each image point is 2× as far from (1, 1)

..... [3]

(d) A translation.

Translation + column vector
e.g. Translation by $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$

..... [2]

Question 34

Vectors, magnitude and vector geometry

a = (3, −1) and **b** = (−2, 5) as column vectors.

(a) Find 2**a** + **b**.

$$2\begin{pmatrix}3\\-1\end{pmatrix}+\begin{pmatrix}-2\\5\end{pmatrix}=\begin{pmatrix}6\\-2\end{pmatrix}+\begin{pmatrix}-2\\5\end{pmatrix}$$

$\begin{pmatrix}4\\3\end{pmatrix}$ [2]

(b) Find |2**a** + **b**|.

$$\sqrt{(4^2+3^2)}=\sqrt{25}$$

5 [2]

(c) OA = **a**, OB = **b**, M is the midpoint of AB. Find AB and OM in terms of **a** and **b**.

$$AB = AO + OB = -a + b$$

$$OM = OA + \frac{1}{2}AB = a + \frac{1}{2}(b - a)$$

$AB = b - a$, $OM = \frac{1}{2}(a + b)$ [3]

8

Probability

E8.1 - E8.4 · 2 questions · 0:24 on the clock

Question 35

Relative and expected frequency

The probability that a machine produces a faulty component is 0.15.

(a) Find the expected number of faulty components in a batch of 400.

$expected = n \times p = 400 \times 0.15$

60 [1]

(b) In a trial, 72 out of 400 were faulty. Find the relative frequency and comment.

$\frac{72}{400} = 0.18 > 0.15$

more faults than expected — but one trial only,
so repeat with more trials

0.18 [2]

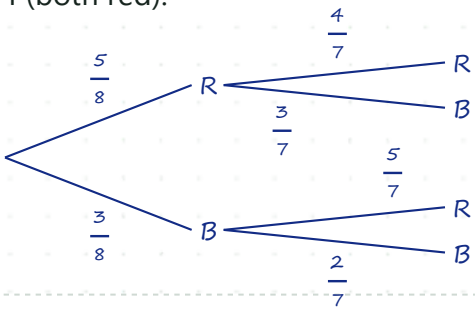
Question 36

Tree diagrams and conditional probability

A bag holds 5 red and 3 blue counters. Two are taken at random **without replacement**.

*AND → multiply along branches · OR → add the outcomes · without replacement changes **both** numerator and denominator*

(a) P(both red).



$\frac{5}{8} \times \frac{4}{7} = \frac{20}{56}$

$\frac{5}{14}$ [2]

(b) P(one of each colour).

RB or BR: $2 \times \frac{5}{8} \times \frac{3}{7} = \frac{30}{56}$

$\frac{15}{28}$ [3]

Question 36

Tree diagrams and conditional probability

A bag holds 5 red and 3 blue counters. Two are taken at random **without replacement**.

(c) P(second is red, given the first was red).

after one red: 4 red left out of 7

$\frac{4}{7}$ [1]

9

Statistics

E9.1 - E9.7 · 3 questions · 0:42 on the clock

Question 37

Averages and measures of spread

The table shows the time t minutes taken by 30 students to finish a task.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	4	11	9	6

(a) Calculate an estimate of the mean.

midpoints 5, 15, 25, 35
$\Sigma fx = 20 + 165 + 225 + 210 = 620$
$620 \div 30 = 20.67$

20.7 min [4]

(b) Write down the modal class.

highest frequency (11)

$10 < t \leq 20$ [1]

Question 37

Averages and measures of spread

The table shows the time t minutes taken by 30 students to finish a task.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	4	11	9	6

(c) Identify the class containing the median.

cumulative: 4, 15, 24, 30
median = 15.5th value → past the 15th

$20 < t \leq 30$ [1]

Question 38

Cumulative frequency diagrams

A cumulative frequency curve is drawn for 80 values.

*Plot points at the **upper** class boundary and join with a smooth curve*

(a) At what value of the cumulative frequency do you read the median, the lower quartile and the upper quartile?

median: $\frac{1}{2} \times 80 = 40$

LQ: $\frac{1}{4} \times 80 = 20$, UQ: $\frac{3}{4} \times 80 = 60$

40, 20, 60 [3]

(b) If LQ = 24 and UQ = 41, find the interquartile range.

$IQR = UQ - LQ = 41 - 24$

17 [1]

Question 39

Histograms

A histogram has unequal class widths.

*frequency density = frequency ÷ class width — the vertical axis is **always** labelled Frequency density*

(a) The class $20 < x \leq 50$ has frequency 24. Find the frequency density.

$$fd = \frac{\text{frequency}}{\text{class width}} = \frac{24}{30}$$

0.8 [2]

(b) Another bar has frequency density 2.4 over the class $50 < x \leq 60$. Find its frequency.

$$\text{frequency} = fd \times \text{class width} = 2.4 \times 10$$

24 [2]

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Answer key 1 of 4

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
01	**E1.1**	Types of number, HCF and LCM	$168 = 2^3 \times 3 \times 7$ $180 = 2^2 \times 3^2 \times 5$ | HCF $= 2^2 \times 3 =$ **12** LCM $= 2^3 \times 3^2 \times 5 \times 7 =$ **2520**
02	**E1.2**	Sets and Venn diagrams	$n(P \cup C) = 40 - 7 = 33$, so $24 + 19 - x = 33 \rightarrow$ **$n(P \cap C) = 10$**; $n(P \cap C') = 24 - 10 =$ **14**
03	**E1.3 / E1.7**	Powers, roots and the index laws	(a) $27^{-2/3} = 1 \div (\sqrt[3]{27})^2 = 1 \div 9 =$ **1/9** | (b) $16a^{12}b^4 \div 4a^5b^2 =$ **$4a^7b^2$**
04	**E1.4**	Fractions, decimals and recurring decimals	(a) $0.4\dot{2} = 42/99 =$ **14/33** | (b) $7/3 - 7/4 = (28 - 21)/12 =$ **7/12**
05	**E1.8**	Standard form	(a) $0.4 \times 10^7 =$ **4×10^6** | (b) $60\,000 + 450\,000 = 510\,000 =$ **5.1×10^5**
06	**E1.10**	Limits of accuracy (bounds)	$12.35 \le L < 12.45$, $8.55 \le W < 8.65$ | (b) $12.45 \times 8.65 =$ **107.6925 cm²** | (c) $2(12.35 + 8.55) =$ **41.8 cm**
07	**E1.11**	Ratio and proportion	$7 - 3 = 4$ parts = \$84, so 1 part = \$21. Total $= 15 \times 21 =$ **\$315**
08	**E1.13**	Percentages and reverse percentages	(a) $161 \div 1.15 =$ **\$140** (not 161×0.85) | (b) $2000 \times 1.035^5 =$ **\$2375.37**
09	**E1.17**	Exponential growth and decay	$18\,000 \times 0.88^4 =$ **\$10 795** (10 794.55)
10	**E1.18**	Surds	(a) $5\sqrt{2} + 3\sqrt{2} =$ **$8\sqrt{2}$** | (b) $6\sqrt{3}/3 =$ **$2\sqrt{3}$** | (c) $\times (2 - \sqrt{3})/(2 - \sqrt{3}) \rightarrow (2 - \sqrt{3})/1 =$ **$2 - \sqrt{3}$**

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Answer key 2 of 4

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
11	E2.2	Expanding and factorising	(a) **$2x^2 + 7x - 15$** | (b) **$(2x - 3)(3x + 4)$** | (c) **$(3x - 7y)(3x + 7y)$** | (d) **$(a - 2b)(x + y)$**
12	E2.3	Algebraic fractions	(a) $(x - 3)(x + 3) / (x + 3)(x + 4) =$ **$(x - 3)/(x + 4)$** | (b) **$(5x - 4) / (x + 1)(x - 2)$**
13	E2.5	Equations and rearranging	(a) **$x = 4, y = 2$** | (b) $x = (7 \pm \sqrt{73})/4 =$ **3.89 or −0.39** | (c) **$r = \sqrt[3]{(3V / 4\pi)}$** | (d) **$(x - 3)^2 - 7$**
14	E2.5	Non-linear simultaneous equations	$x^2 - 3x - 4 = 0 \rightarrow (x - 4)(x + 1) = 0 \rightarrow$ **$x = 4, y = 12$** or **$x = -1, y = -3$**
15	E2.6	Inequalities	(a) $-2x < 6 \rightarrow$ **$x > -3$** (open circle) | (b) $-4 \le 2x < 8 \rightarrow$ **$-2 \le x < 4$**
16	E2.7	Sequences	(a) **$4n + 3$** | (b) second difference 2 → **$n^2 + 2n$** | (c) **$5 \times 2^{n-1}$**
17	E2.8	Direct and inverse proportion	(a) **$y = 48/x^2$** | (b) **$y = 3$** | (c) $x^2 = 64 \rightarrow$ **$x = 8$**
18	E2.9	Graphs in practical situations	(b) $15 \div 10 =$ **1.5 m/s²** | (c) area = ½(10)(15) + 30(15) + ½(20)(15) = 75 + 450 + 150 = **675 m**
19	E2.11	Sketching curves	(a) $(x - 3)^2 - 4$, turning point **(3, −4)**, roots **$x = 1$ and $x = 5$** | (b) asymptotes **$x = 0$ and $y = 0$**
20	E2.12	Differentiation	(a) **$3x^2 - 12x$** | (b) $3x(x - 4) = 0 \rightarrow$ **(0, 5)** and **(4, −27)** | (c) $d^2y/dx^2 = 6x - 12$: at $x = 0$ it is $-12 \rightarrow$ **maximum**; at $x = 4$ it is $12 \rightarrow$ **minimum**

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Answer key 3 of 4

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	CODE	TOPIC	ANSWER
21	E2.13	Functions	(a) f(4) = **8** | (b) g(2) = **4** | (c) **(x + 4)/3** | (d) 3(3x − 4) − 4 = **9x − 16**
22	E3.1 - E3.7	All of coordinate geometry in one pair of points	(a) **(2, 4)** | (b) √(8²+6²) = **10** | (c) **¾** | (d) **y = ¾x + 2.5** | (e) **y = ¾x** | (f) gradient −4/3 → **y = −(4/3)x + 15**
23	E4.1 / E4.6	Geometrical terms and angles	(a) exterior = 360 ÷ 12 = **30°**; interior = **150°**; sum = (12 − 2) × 180 = **1800°** | (b) corresponding, alternate, co-interior (sum 180°)
24	E4.4 / E4.5	Similarity and symmetry	k = 1.5 | (a) 32 × 1.5³ = 32 × 3.375 = **108 cm³** | (b) 20 × 1.5² = **45 cm²**
25	E4.7 / E4.8	Circle theorems	(a) **65°** (angle at centre is twice the angle at the circumference) | (b) **115°** (cyclic quadrilateral) | (c) **58°** (alternate segment) | (d) √(289 − 64) = **15 cm** (tangent ⊥ radius)
26	E5.2 / E5.5	Area, perimeter and compound shapes	(a) ½(7 + 11)(6) = **54 cm²** | (b) 60 − ½π(3)² = 60 − 14.14 = **45.9 cm²**
27	E5.3	Circles, arcs and sectors	(a) (140/360)(2π × 9) = **22.0 cm** | (b) (140/360)(π × 81) = **99.0 cm²** | (c) 21.99 + 9 + 9 = **40.0 cm**
28	E5.4	Surface area and volume	(a) l = 13 cm; total SA = π(5)(13) + π(5)² = 90π = **283 cm²**; V = ⅓π(25)(12) = 100π = **314 cm³** | (b) V = 288π = **905 cm³**; SA = 144π = **452 cm²**
29	E6.1 / E6.2	Pythagoras and right-angled trigonometry	(a) sin θ = 7/12 → **θ = 35.7°** | (b) √(144 − 49) = √95 = **9.75 cm**
30	E6.4	Trigonometric graphs and equations	(a) **x = 30° and 150°** | (b) cos x = −0.5 → **x = 120° and 240°**

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Answer key 4 of 4

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	CODE	TOPIC	ANSWER
31	E6.5	Sine rule, cosine rule and area	(a) 8.4 sin63° ÷ sin47° = **10.2 cm** | (b) a^2 = 64 + 121 − 176 cos47° = 64.97 → **8.06 cm** | (c) ½(8)(11) sin47° = **32.2 cm²**
32	E6.6	Pythagoras and trigonometry in 3D	(a) √(36 + 64 + 100) = √200 = **14.1 cm** | (b) base diagonal = 10, tan θ = 10/10 → **θ = 45°**
33	E7.1	Transformations	A negative scale factor puts the image on the **opposite side of the centre and inverted**; a fractional scale factor makes it **smaller**. One word per transformation — naming two transformations loses the mark.
34	E7.2 - E7.4	Vectors, magnitude and vector geometry	(a) **(4, 3)** | (b) √(16+9) = **5** | (c) *AB* = **b − a**; *OM* = **a** + ½(**b** − **a**) = **½(a + b)**
35	E8.2	Relative and expected frequency	(a) 0.15 × 400 = **60** | (b) 72/400 = **0.18** — higher than expected, though a single trial is weak evidence
36	E8.3 / E8.4	Tree diagrams and conditional probability	(a) 5/8 × 4/7 = **5/14** | (b) 2 × (5/8 × 3/7) = **15/28** | (c) **4/7** — the condition changes the denominator too
37	E9.3	Averages and measures of spread	(a) Σfx = 20 + 165 + 225 + 210 = 620; 620 ÷ 30 = **20.7 min** | (b) **10 < t ≤ 20** | (c) 15th/16th value → **20 < t ≤ 30**
38	E9.6	Cumulative frequency diagrams	(a) median at **40**, LQ at **20**, UQ at **60** | (b) IQR = 41 − 24 = **17**
39	E9.7	Histograms	(a) 24 ÷ 30 = **0.8** | (b) 2.4 × 10 = **24**